

DATA MINING: DATA

What is Data?

- Collection of *data objects*- *data sets*
 - Object is also known as record, point, case, sample, entity, or instance
- Data objects are described by a number of attributes
- An **attribute** is a property or characteristic of an object that may vary either from one object to another or from one time to another
 - Examples: eye color of a person, temperature, etc.
 - Attribute is also known as variable, field, characteristic, dimension, or feature

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

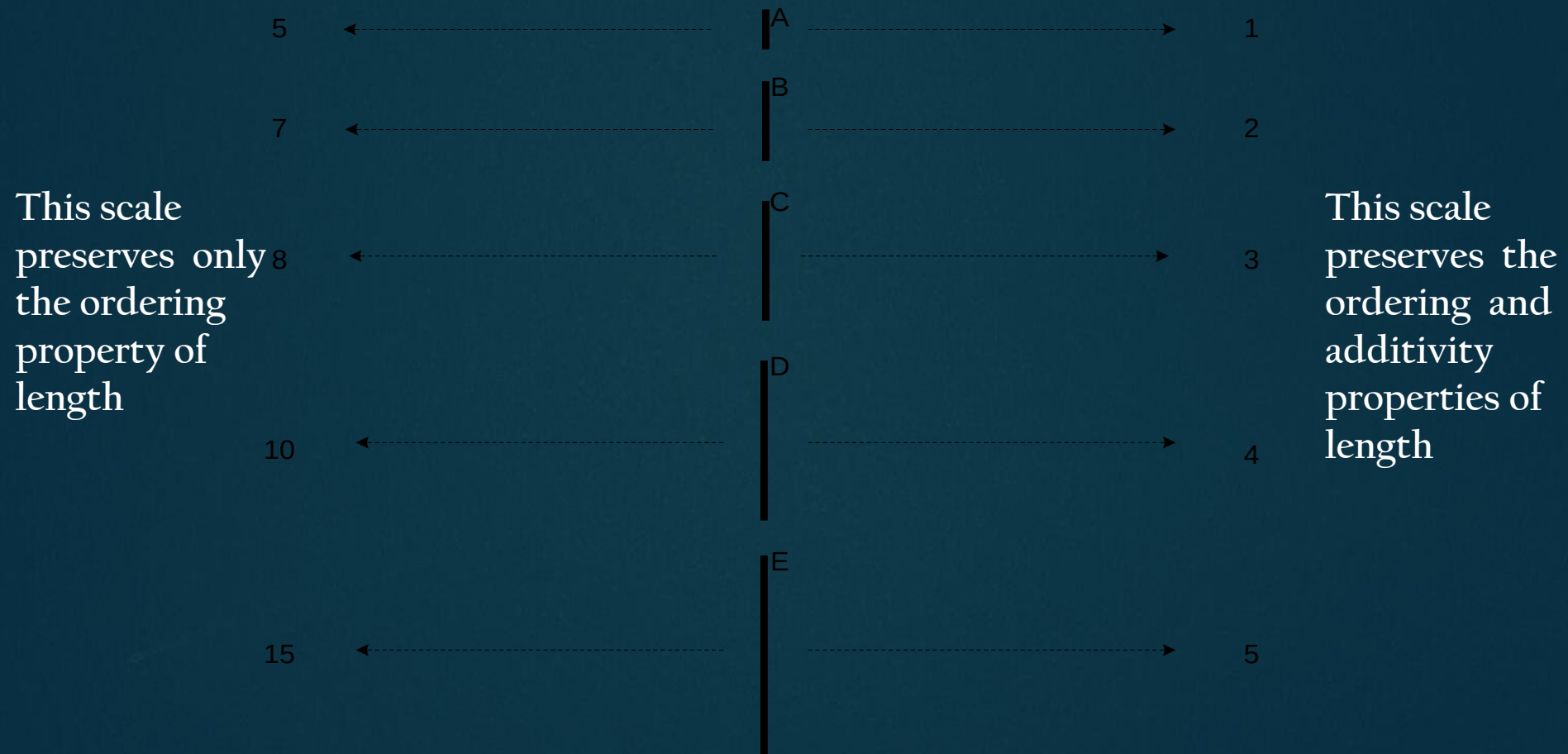
- A measurement scale is a rule (function) that associates a numerical or symbolic value with an attribute of an object
- Process of measurement is the application of a measurement scale to associate a value with a particular attribute of a specific object

Attribute Values

- **Attribute values** are numbers or symbols assigned to an attribute for a particular object
- **Distinction between attributes and attribute values**
 - Same attribute can be mapped to different attribute values
 - Example: height can be measured in feet or meters
 - Different attributes can be mapped to the same set of values
 - Example: Attribute values for **ID** and **age** are **integers**
 - But properties of attribute values can be different
 - (Average of ID is meaningless)
- **Properties of an attribute need not be the same as the properties of the values used to measure it**

Measurement of Length

- The way you measure an attribute may not match the attributes properties.



Types of Attributes

There are different types of attributes

- Nominal
- Ordinal
- Interval
- Ratio

Types of Attributes

Nominal

- Values of a nominal attribute are just different names
- Nominal values provide only enough information to distinguish one object from another ($=$, $\neq$)
- Examples: ID numbers, eye color, zip codes

Types of Attributes

Ordinal

- The values of an ordinal attribute provide enough information to order objects($\langle, \rangle$)
- Ordinal Attributes contains values that have a meaningful sequence or ranking(order) between them
- The magnitude between values is not actually known- the order of values shows what is important but don't indicate how important it is
- Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height {tall, medium, short}

Types of Attributes

Interval

- The differences between values are meaningful(+,-)
- A unit of measurement exists
- Intervals between each value are equally split
- No true “zero value” (0 deg Celsius doesn't mean “No temperature”)
- Examples: calendar dates, temperature in Celsius or Fahrenheit
- If temperature in Delhi is 40 ° Celsius and that in Shimla is 20 ° Celsius, then Delhi is 20 ° Celsius hotter than Shimla (taking difference)
- But you cannot say Delhi is twice as hot as Shimla

Types of Attributes

Ratio

- Meaningful ratio can be constructed(/,*)
- There is an absolute zero
- Examples: monetary quantities, counts, age, mass, length, electrical current, temperature in Kelvin
- Examples:
 - 40 kg is twice as heavy as 20 kg
 - No. of clients may be 0 in last six months or may be twice as that of previous six months

Types of Attributes

Categorical or Qualitative attributes: Nominal and Ordinal attributes

- Attribute where the values correspond to discrete categories- Deals with characteristics and descriptors that can't be easily measured, but can be observed subjectively
- Lack most of the properties of numbers
- Even if they are represented by numbers, i.e., integers, they should be treated more like symbols

Types of Attributes

Quantitative or numeric attributes-Interval and Ratio attributes

- Represented by numbers
- Have most of the properties of numbers
- Can be integer-valued or continuous

Properties of Attribute Values

- The type of an attribute depends on which of the following properties/operations it possesses:
 - Distinctness: $=$ $+$
 - Order: $<$ $>$
 - Differences $+$ $-$
 - Ratios $*$ $/$
- Nominal attribute: distinctness
- Ordinal attribute: distinctness & order
- Interval attribute: distinctness, order & meaningful differences
- Ratio attribute: all 4 properties/operations

- Is it physically meaningful to say that a temperature of 10° is twice that of 5° on
 - the Celsius scale?
 - the Fahrenheit scale?
 - the Kelvin scale?
- Consider measuring the height above average
 - If Bill's height is three inches above average and Bob's height is six inches above average, then would we say that Bob is twice as tall as Bill?
 - Is this situation analogous to that of temperature?

		Attribute Type	Transformation	Comments
Categorical Qualitative		Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
		Ordinal	An order preserving change of values, i.e., $new_value = f(old_value)$ where f is a monotonic function	An attribute encompassing the notion of good, better best can be represented equally well by the values {1, 2, 3} or by { 0.5, 1, 10}.
Numeric Quantitative		Interval	$new_value = a * old_value + b$ where a and b are constants	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
		Ratio	$new_value = a * old_value$	Length can be measured in meters or feet.

This categorization of attributes is due to S. S. Stevens

Describing Attributes by the Number of Values

Discrete Attribute

- Has only a finite or countably infinite set of values
- Examples: zip codes, counts, or the set of words in a collection of documents
- Often represented as integer variables
- **Binary attributes**
 - Special case of discrete attributes
 - Assume only two values, e.g., true/false, yes/no, male/female, or 0/1
 - Often represented as Boolean variables, or as integer variables that only take the values 0 or 1

Continuous Attribute

- Has real numbers as attribute values
- Examples: temperature, height, or weight.
- Practically, real values can only be measured and represented using a finite number of digits
- Continuous attributes are typically represented as floating-point variables

Asymmetric Attributes

- Only presence (a non-zero attribute value) is regarded as important
- Binary attributes where only non-zero values are important are called asymmetric binary attributes
- A binary variable is said to be symmetric if both of its states are of the same value and carry the same weight, and there is no preference
- Example: attribute of gender
- An asymmetric binary attribute is one in which outcomes are not of equal importance

Important Characteristics of Data sets

Dimensionality

- Number of attributes that the objects in the data set possess
- High dimensional data brings a number of challenges
- Difficulties associated with analyzing high-dimensional data-

Curse of dimensionality

Sparsity

- For some data sets, most attributes of an object have values of 0
- Advantage because usually only the non-zero values need to be stored and manipulated
- Results in significant savings with respect to computation time and storage
- Some data mining algorithms work well only for sparse data

Resolution

- It is possible to obtain data at different levels of resolution
- Properties of the data are different at different resolutions
- Patterns in the data depend on the level of resolution
- If the resolution is too fine, a pattern may not be visible or may be buried in noise; if the resolution is too coarse, the pattern may disappear
- Example: Variations in atmospheric pressure on a scale of hours reflect the movement of storms and other weather systems. On a scale of months, such phenomena are not detectable

Types of data sets

- Record
 - Data Matrix
 - Document Data
 - Transaction Data
- Graph
 - World Wide Web
 - Molecular Structures
- Ordered
 - Spatial Data
 - Temporal Data
 - Sequential Data
 - Genetic Sequence Data

Record Data

- Data set is a collection of records(data objects), which consists of a fixed set of data fields (attributes)
- Record data is usually stored either in flat files or in relational databases

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
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Transaction Data(Market Basket Data)

- A special type of record data, where each record (transaction) involves a set of items
- Example: Grocery store (Market basket) data
 - The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items

<i>TID</i>	<i>Items</i>
1	Bread , Coke , Milk
2	Bread , Bread
3	Bread , Coke , Diaper , Milk
4	Bread , Bread , Diaper , Milk
5	Coke , Diaper , Milk

Data Matrix

- Data objects have the same fixed set of numeric attributes
- Such a data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute
- Because it consists of numeric attributes, standard matrix operation can be applied to transform and manipulate the data

Projection of x Load	Projection of y Load	Distance	Load	Thickness
10.23	5.27	15.22	27	1.2
12.65	6.25	16.22	22	1.1
13.54	7.23	17.34	23	1.2
14.27	8.43	18.45	25	0.9

Sparse Data Matrix

- Special case of a data matrix in which the attributes are of the same type and are asymmetric; i.e., only non-zero values are important
- Example: Transaction data

TID	Bread	Milk	Diapers	Beer	Eggs	Cola
1	1	1	0	0	0	0
2	1	0	1	1	1	0
3	0	1	1	1	0	1
4	1	1	1	1	0	0
5	1	1	1	0	0	1

Example: Document Data

- Each document becomes a 'term' vector
- Each term is a component (attribute) of the vector
- The value of each component is the number of times the corresponding term occurs in the document

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Graph based data

- A graph can sometimes be a convenient and powerful representation for data
- Types:
 - Graph capturing relationships among data objects
 - Data objects themselves are represented as graph

Data with Relationships among Objects

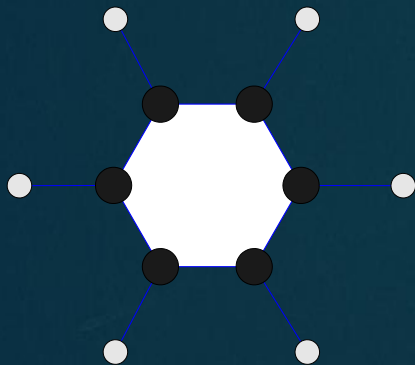
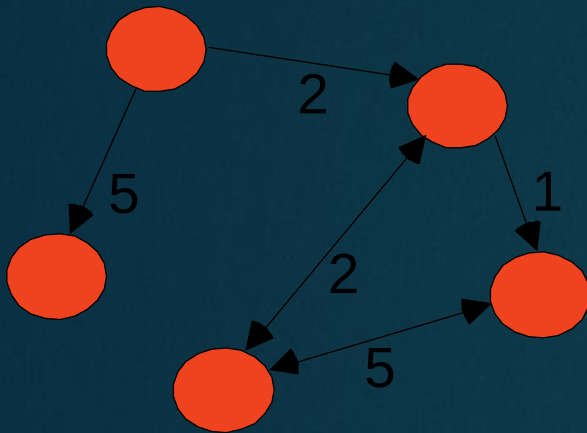
- Relationships among objects frequently convey important information
- Data **objects** are mapped to **nodes** of the graph
- The **relationships among objects** are captured by the **links between objects and link properties** (such as direction and weight)
- Example: Web pages on the World Wide Web

Data with Objects That Are Graphs

- If objects contain subobjects that have relationships, then such objects are frequently represented as graphs
- structure of chemical compounds can be represented by a graph, where the nodes are atoms and the links between nodes are chemical bonds
- Graph representation makes it possible to
 - determine which substructures occur frequently in a set of compounds
 - ascertain whether the presence of any of these substructures is associated with the presence or absence of certain chemical properties, such as melting point or heat of formation.

Graph Data

- Examples: Generic graph, a molecule, and webpages



Useful Links:

- [Bibliography](#)
- Other Useful Web sites
 - [ACM SIGKDD](#)
 - [KDnuggets](#)
 - [The Data Mine](#)

Knowledge Discovery and Data Mining Bibliography

(Gets updated frequently, so visit often!)

- [Books](#)
- [General Data Mining](#)

Book References in Data Mining and Knowledge Discovery

Usama Fayyad, Gregory Piatetsky-Shapiro, Padhraic Smyth, and Ramasamy uthurasamy, "Advances in Knowledge Discovery and Data Mining", AAAI Press/the MIT Press, 1996.

J. Ross Quinlan, "C4.5: Programs for Machine Learning", Morgan Kaufmann Publishers, 1993.
Michael Berry and Gordon Linoff, "Data Mining Techniques (For Marketing, Sales, and Customer Support)", John Wiley & Sons, 1997.

General Data Mining

Usama Fayyad, "Mining Databases: Towards Algorithms for Knowledge Discovery", Bulletin of the IEEE Computer Society Technical Committee on data Engineering, vol. 21, no. 1, March 1998.

Christopher Matheus, Philip Chan, and Gregory Piatetsky-Shapiro, "Systems for knowledge Discovery in databases", IEEE Transactions on Knowledge and Data Engineering, 5(6):903-913, December 1993.

Benzene Molecule: C₆H₆

Ordered Data

- Attributes may have relationships that involve order in time or space

Sequential Data/Temporal data

- Extension of record data, where each record has a time associated with it
- Example: retail transaction data set that also stores the time at which the transaction took place

Time	Customer	Items Purchased
t1	C1	A, B
t2	C3	A, C
t2	C1	C, D
t3	C2	A, D
t4	C2	E
t5	C1	A, E

Customer	Time and Items Purchased
C1	(t1: A,B) (t2:C,D) (t5:A,E)
C2	(t3: A, D) (t4: E)
C3	(t2: A, C)

Ordered Data

Sequence Data

- Consists of a data set that is a sequence of individual entities, such as a sequence of words or letters
- Quite similar to sequential data, except that there are no time stamps; instead, there are positions in an ordered sequence

Ordered Data

- Example: Genomic sequence data

```
GGTTCCGCCTTCAGCCCCGCGCC  
CGCAGGGGCCCGCCCCGCGCCGTC  
GAGAAGGGCCCCGCCTGGCGGGCG  
GGGGGAGGCGGGGGCCGCCCGAGC  
CCAACCGAGTCCGACCAGGTGCC  
CCCTCTGCTCGGCCTAGACCTGA  
GCTCATTAGGCGGCAGCGGACAG  
GCCAAGTAGAACACGCGAAGCGC  
TGGGCTGCCTGCTGCGACCAGGG
```

Ordered Data

Time Series Data

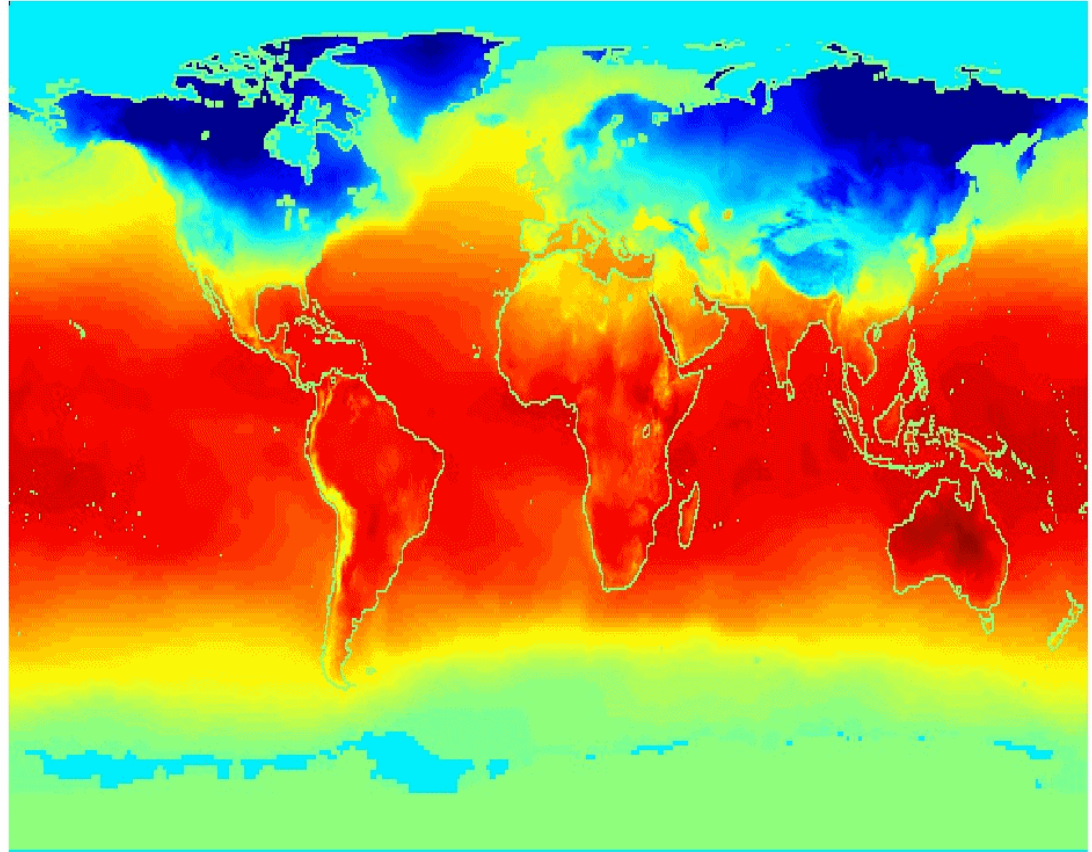
- Time-series data contain values that are typically generated by continuous measurement over time
- If two measurements are close in time, then the values of those measurements are often very similar
- Financial data set might contain objects that are time series of the daily prices of various stocks



Ordered Data

**Average
Monthly
Temperature**

Jan



Ordered Data

- Spatial Data
 - Some objects have spatial attributes, such as positions or areas
 - Examples: Weather data (precipitation, temperature, pressure) that is collected for a variety of geographical locations
 - Objects that are physically close tend to be similar in other ways also
 - Example: Data sets that are the result of measurements or model output taken at regularly or irregularly distributed points on a two- or three-dimensional grid or mesh